AI Builder Path

RAG SYSTEM DESIGN

Architecture & Workflow

Presented by Aswin M A

SYSTEM OVERVIEW

Objective

To design an intelligent chatbot that answers employee queries based strictly on the Company HR & IT Policy document, eliminating manual searching and ensuring 24/7 support accuracy

key logic

The system uses RAG. Instead of relying on a model's broad knowledge, we provide the model with specific policy "chunks" to ground its answers in actual company data.

DATA INGESTION

Document Collection
The system collects organizational knowledge from internal documents such as HR Policy Documents and IT Policy Documents

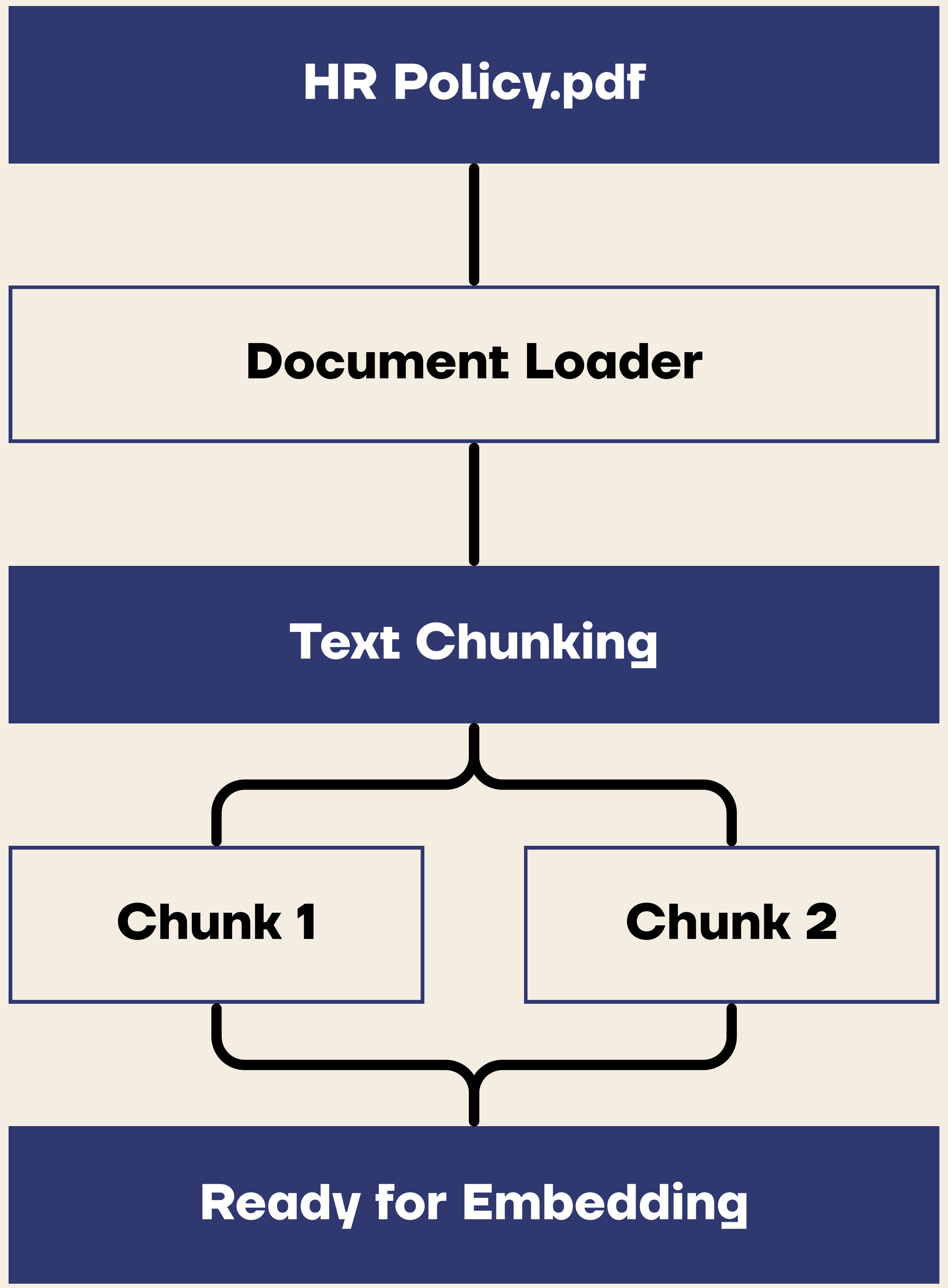
Document Loading
Documents are loaded into the RAG pipeline for processing and indexing.

Text Chunking: Large documents are divided into smaller chunks to improve retrieval accuracy and maintain contextual relevance.

- Chunk Size: 300 Characters

Why Chunking is Important?

- Improves retrieval performance
- Reduces context loss
- Enables efficient vector search
- Provides more accurate responses



EMBEDDING MODEL

The embedding model converts text chunks into high-dimensional numerical vectors that capture the semantic meaning of the content rather than just the exact words.

By representing text as vectors, the system can identify relationships between concepts and retrieve relevant information even when different wording is used.

VPN Access — **Embedding Model** — **[0.23, -0.67, 0.91, ...]**

VECTOR DATABASE

Why FAISS?

FAISS enables efficient semantic search by comparing embeddings instead of exact keywords.

Benefits

- Fast retrieval of relevant policy information
- Improves answer accuracy in RAG systems

Storage

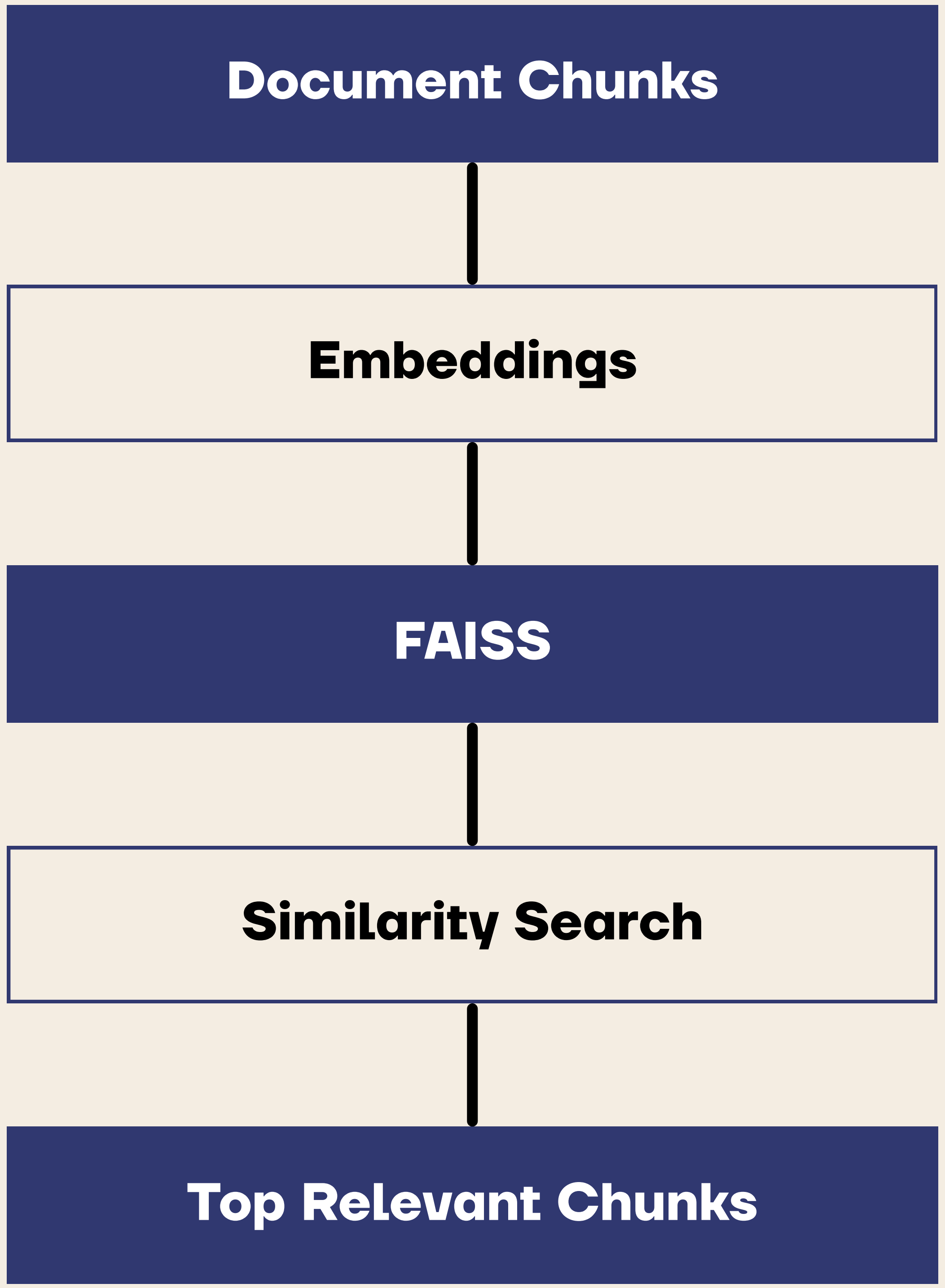
Using FAISS to store and index document embeddings for fast retrieval.

Similarity Search

Matches employee questions with semantically similar policy chunks.

Persistence

Stores vector indexes for reuse without rebuilding embeddings.



INTELLIGENT RETRIEVAL

Context Retrieval

When a user submits a question, the system identifies and retrieves the most relevant information from the knowledge base before generating a response.

Retrieval Process

- The user's question is converted into an embedding vector.
- The vector database performs semantic similarity search.
- The most relevant document chunks are retrieved.
- Retrieved context is combined and sent to the Large Language Model (LLM).

Example

User Query:

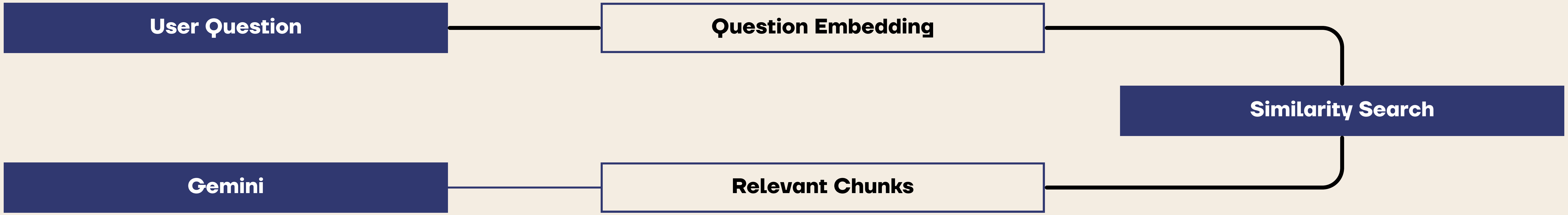
"How many leaves do employees receive per year?"

Retrieved Context:

"Employees are entitled to 12 casual leaves annually as per company policy."

LLM Response:

"Employees receive 12 casual leaves per year according to the HR policy."



BENEFITS OF RAG SYSTEM

Accuracy

- Responses are grounded in company documents.
- Reduces hallucinations.

Efficiency

- Employees find information quickly.
- Reduces manual document searching.

Maintainability

Updating documents automatically updates the knowledge base after re-indexing.

Scalability

- Supports large collections of documents.
- Easy to add new policies and knowledge sources.

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THANK YOU

Thanks For Watching

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